

Scientific Computing

Mathematical Foundations for Machine Learning

Course Conspect

Course Structure

Block I: Probability and Statistics

Lecture	Topic
1	Probability models, conditioning, and random variables
2	Distributions, expectation, and variance
3	Joint distributions, dependence, and random vectors
4	Samples, concentration, and limit theorems
5	Statistical estimation: method of moments, maximum likelihood, and MAP
6	Sampling distributions, confidence intervals, and bootstrap
7	Entropy, cross-entropy, KL divergence, Fisher information, and the Cramér–Rao bound

Block II: Linear Algebra, Calculus, and Optimization

Lecture	Topic
8	Geometry of data, projections, and least squares
9	Eigenvalues, positive-definite matrices, singular value decomposition, and PCA
10	Multivariable calculus, matrix calculus, and automatic differentiation
11	Optimization for machine learning

Block III: Numerical Computing and Autodiff

Lecture	Topic
12	NumPy, visualization, and numerical linear algebra
13	PyTorch, autograd, and the integrated capstone

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Chapter 1

Probability Models, Conditioning, and Random Variables

Probability starts with a model: a precise description of what may happen, which statements about the outcome are meaningful, and how probabilities are assigned. Conditioning changes the reference population, while a random variable turns outcomes into numerical values whose distribution can be studied.

1.1 Probability spaces and events

Definition 1.1 (Probability space). A probability space is a triple

$$(\Omega, \mathcal{F}, \mathbb{P}).$$

Here Ω is the *sample space*, whose elements are outcomes; $\mathcal{F}$ is a collection of subsets of Ω , called events; and $\mathbb{P}$ assigns a probability to every event.

The event collection $\mathcal{F}$ is closed under complements and countable unions:

$$A \in \mathcal{F} \implies A^c \in \mathcal{F}, \quad A_1, A_2, \dots \in \mathcal{F} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}.$$

It also contains Ω . In a finite model one normally takes $\mathcal{F} = 2^\Omega$, the set of all subsets of Ω .

For events $A, B \subseteq \Omega$:

- $A \cap B$ means that both events occur;
- $A \cup B$ means that at least one occurs;
- $A^c = \Omega \setminus A$ means that A does not occur;
- $A \setminus B = A \cap B^c$ means that A occurs but B does not.

Definition 1.2 (Probability axioms). For events in $\mathcal{F}$, a probability measure satisfies:

$$\begin{aligned} \mathbb{P}(A) &\geq 0, \\ \mathbb{P}(\Omega) &= 1, \\ \mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) &= \sum_{i=1}^{\infty} \mathbb{P}(A_i) \quad \text{for pairwise disjoint } A_i. \end{aligned}$$

Several basic rules follow from these axioms:

$$\mathbb{P}(\emptyset) = 0, \quad (1.1)$$

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \quad (1.2)$$

$$A \subseteq B \implies \mathbb{P}(A) \leq \mathbb{P}(B), \quad (1.3)$$

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B). \quad (1.4)$$

The last identity is the inclusion–exclusion rule. Adding $\mathbb{P}(A)$ and $\mathbb{P}(B)$ counts the overlap twice, so it must be subtracted once.

Example 1.3 (Two fair coin tosses). Toss a fair coin twice and record the ordered result. Then

$$\Omega = \{HH, HT, TH, TT\}, \quad \mathbb{P}(\{\omega\}) = \frac{1}{4} \quad \text{for every } \omega \in \Omega.$$

Let

$$A = \{\text{first toss is } H\} = \{HH, HT\}, \quad B = \{\text{second toss is } H\} = \{HH, TH\}.$$

Then

$$A \cap B = \{HH\}, \quad A \cup B = \{HH, HT, TH\}, \quad A^c = \{TH, TT\}.$$

Inclusion–exclusion gives

$$\mathbb{P}(A \cup B) = \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}.$$

For a finite sample space with equally likely outcomes,

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$

This is a counting shortcut, not an additional axiom: it is valid only after equal likelihood has been justified.

1.2 Conditional probability and total probability

Definition 1.4 (Conditional probability). For $\mathbb{P}(B) > 0$, the probability of A given B is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \quad (1.5)$$

Conditioning restricts the reference set to B and renormalizes the remaining probabilities. The event after the vertical bar determines the denominator. In general, $\mathbb{P}(A \mid B)$ and $\mathbb{P}(B \mid A)$ are different quantities.

Rearranging (1.5) gives the multiplication rule

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A). \quad (1.6)$$

Example 1.5 (At least one head). In the two-toss model, let

$$E = \{\text{at least one head}\} = \{HH, HT, TH\}.$$

Given E , the probability that the second toss is heads is

$$\mathbb{P}(B \mid E) = \frac{\mathbb{P}(B \cap E)}{\mathbb{P}(E)} = \frac{2/4}{3/4} = \frac{2}{3}.$$

The unconditional probability is $\mathbb{P}(B) = 1/2$. Conditioning removes the outcome TT , in which B fails.

Theorem 1.6 (Law of total probability). *Let $B_1, \dots, B_m$ be a partition of Ω : the events are pairwise disjoint and their union is Ω . If $\mathbb{P}(B_i) > 0$, then*

$$\mathbb{P}(A) = \sum_{i=1}^m \mathbb{P}(A \mid B_i) \mathbb{P}(B_i). \quad (1.7)$$

Indeed, the events $A \cap B_i$ are disjoint and their union is A . Add their probabilities and then apply the multiplication rule. For a binary partition B, B^c ,

$$\mathbb{P}(A) = \mathbb{P}(A \mid B) \mathbb{P}(B) + \mathbb{P}(A \mid B^c) \mathbb{P}(B^c).$$

1.3 Independence and mutual exclusivity

Definition 1.7 (Independence). Events A and B are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B). \quad (1.8)$$

If $\mathbb{P}(B) > 0$, this is equivalent to $\mathbb{P}(A \mid B) = \mathbb{P}(A)$.

Independence means that learning whether one event occurred does not change the probability of the other. In the coin example,

$$\mathbb{P}(A \cap B) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2},$$

so the events “first toss is heads” and “second toss is heads” are independent.

Definition 1.8 (Mutual exclusivity). Events A and B are mutually exclusive, or disjoint, if

$$A \cap B = \emptyset.$$

They cannot occur together.

Proposition 1.9. *Two mutually exclusive events with positive probability are not independent.*

Proof. Mutual exclusivity gives $\mathbb{P}(A \cap B) = 0$, while positive probabilities give $\mathbb{P}(A) \mathbb{P}(B) > 0$. Thus (1.8) cannot hold. $\square$

The distinction is essential:

	Mutual exclusivity	Independence
Can both occur?	No	Usually yes
Defining condition	$A \cap B = \emptyset$	$\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B)$
Information meaning	One rules out the other	One does not change the other

1.4 Bayes' theorem and base rates

Theorem 1.10 (Bayes' theorem). *Let $B_1, \dots, B_m$ be a partition of Ω . For $\mathbb{P}(A) > 0$,*

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j) \mathbb{P}(B_j)}{\sum_{i=1}^m \mathbb{P}(A \mid B_i) \mathbb{P}(B_i)}. \quad (1.9)$$

The numerator is the probability of the route B_j followed by A . The denominator adds every mutually exclusive route that can produce A .

Example 1.11 (Rare-defect classifier). Let D mean that an item is defective and $+$ mean that a classifier produces a positive flag. Assume

$$\mathbb{P}(D) = 0.01, \quad \mathbb{P}(+ | D) = 0.99, \quad \mathbb{P}(+ | D^c) = 0.05.$$

The probability of a positive result is

$$\mathbb{P}(+) = 0.99(0.01) + 0.05(0.99) = 0.0594.$$

Therefore

$$\mathbb{P}(D | +) = \frac{0.99(0.01)}{0.0594} = \frac{1}{6} \approx 0.167.$$

The same calculation can be expressed with expected counts in a population of 10,000 items:

Actual state	Positive	Negative	Total
Defect D	99	1	100
No defect D^c	495	9,405	9,900
Total	594	9,406	10,000

Only 99 of the 594 positive items are defective. The sensitivity $\mathbb{P}(+ | D) = 0.99$ is not the posterior $\mathbb{P}(D | +)$. The low base rate and the false positives among the much larger non-defective group dominate the denominator.

For a Bayesian classifier with classes $C_1, \dots, C_K$ and observed features x ,

$$\mathbb{P}(C_k | x) = \frac{p(x | C_k)\mathbb{P}(C_k)}{\sum_{j=1}^K p(x | C_j)\mathbb{P}(C_j)}. \quad (1.10)$$

The factors have distinct roles:

- $\mathbb{P}(C_k)$ is the prior or class base rate;
- $p(x | C_k)$ is the class-conditional likelihood;
- $\mathbb{P}(C_k | x)$ is the posterior after observing x .

The maximum a posteriori rule is

$$\hat{C}_{\text{MAP}}(x) = \arg \max_k p(x | C_k)\mathbb{P}(C_k).$$

1.5 Random variables

Definition 1.12 (Random variable). A real-valued random variable is a function

$$X : \Omega \longrightarrow \mathbb{R}$$

such that $\{\omega : X(\omega) \leq x\}$ is an event for every $x \in \mathbb{R}$. In finite examples this event condition is automatic.

The sampled outcome is random; the mapping X is fixed.

Example 1.13 (Number of heads). For two coin tosses, let X be the number of heads. Then

Outcome ω	HH	HT	TH	TT
$X(\omega)$	2	1	1	0

The support $\{0, 1, 2\}$ is the set of possible values of X ; it is not the original sample space.

A *discrete* random variable takes values in a finite or countable set. An *absolutely continuous* random variable assigns probabilities to intervals through a density. Mixed random variables, with both point masses and a continuous part, also exist.

1.6 PMFs, PDFs, and CDFs

1.6.1 Probability mass functions

Definition 1.14 (PMF). For a discrete random variable X , its probability mass function is

$$p_X(x) = \mathbb{P}(X = x), \quad p_X(x) \geq 0, \quad \sum_x p_X(x) = 1.$$

For the number of heads in two fair tosses,

$$p_X(0) = \frac{1}{4}, \quad p_X(1) = \frac{1}{2}, \quad p_X(2) = \frac{1}{4}.$$

The probability of an event is obtained by summing the masses of its values; for example, $\mathbb{P}(X \geq 1) = p_X(1) + p_X(2) = 3/4$.

1.6.2 Cumulative distribution functions

Definition 1.15 (CDF). For every real-valued random variable X , the cumulative distribution function is

$$F_X(x) = \mathbb{P}(X \leq x).$$

Every CDF is nondecreasing and right-continuous, with

$$\lim_{x \rightarrow -\infty} F_X(x) = 0, \quad \lim_{x \rightarrow \infty} F_X(x) = 1.$$

It determines interval probabilities through

$$\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a). \quad (1.11)$$

For the number of heads,

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{4}, & 0 \leq x < 1, \\ \frac{3}{4}, & 1 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$

For a discrete variable, the jump at x equals its point mass:

$$p_X(x) = F_X(x) - F_X(x^-),$$

where $F_X(x^-)$ denotes the left limit.

1.6.3 Probability density functions

Definition 1.16 (PDF). An absolutely continuous random variable Y has a probability density function f_Y satisfying

$$f_Y(y) \geq 0, \quad \int_{-\infty}^{\infty} f_Y(y) dy = 1,$$

and

$$\mathbb{P}(a < Y \leq b) = \int_a^b f_Y(y) dy. \quad (1.12)$$

Warning. A density value is not a point probability. For an absolutely continuous random variable, $\mathbb{P}(Y = y) = 0$ for every single point, and $f_Y(y)$ may be larger than one. Probability is the area under the density.

The CDF and density are related by

$$F_Y(y) = \int_{-\infty}^y f_Y(t) dt, \quad F'_Y(y) = f_Y(y)$$

at points where the derivative exists.

Example 1.17 (Uniform distribution on the unit interval). Let

$$f_Y(y) = \begin{cases} 1, & 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ y, & 0 \leq y \leq 1, \\ 1, & y > 1. \end{cases}$$

Thus

$$\mathbb{P}(0.2 < Y \leq 0.7) = \int_{0.2}^{0.7} 1 dy = F_Y(0.7) - F_Y(0.2) = 0.5.$$

Chapter 2

Distributions, Expectation, and Variance

A distribution describes how a random quantity varies. The useful starting point is a mechanism: waiting for a request, counting successful trials, adding small measurement errors, or sampling from a population whose members differ. Expectation and variance let us compare these mechanisms and explain what happens when their outputs are transformed, added, or mixed.

2.1 Exponential waiting times

Suppose requests arrive at a service, and let T be the waiting time until the next request. A simple model assumes a constant arrival rate $\lambda > 0$. Over a short interval of length h , conditional on still waiting, the chance of an arrival is approximately λh , regardless of how long we have already waited.

Write $S(t) = \mathbb{P}(T > t)$, the *survival function*. The constant-rate assumption gives

$$S(t+h) = S(t)(1 - \lambda h + o(h)),$$

where $o(h)/h \rightarrow 0$ as $h \rightarrow 0$. Consequently,

$$S'(t) = -\lambda S(t), \quad S(0) = 1,$$

whose solution is $S(t) = e^{-\lambda t}$.

Definition 2.1 (Exponential distribution). The notation $T \sim \text{Exp}(\lambda)$ means that T has density

$$f_T(t) = \begin{cases} \lambda e^{-\lambda t}, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

The parameter λ is a *rate*: if t is in minutes, λ is in inverse minutes.

A density assigns probabilities by area:

$$\mathbb{P}(a < T \leq b) = \int_a^b f_T(t) dt, \quad \int_{-\infty}^{\infty} f_T(t) dt = 1.$$

It is not a point probability: $\mathbb{P}(T = t) = 0$ for every fixed t . Density values can exceed one; their integral over an interval cannot.

The *cumulative distribution function* (CDF) of any random variable is $F_X(x) = \mathbb{P}(X \leq x)$. In this example,

$$F_T(t) = \begin{cases} 0, & t < 0, \\ 1 - e^{-\lambda t}, & t \geq 0. \end{cases}$$

Every CDF is nondecreasing and right-continuous, tends to zero at $-\infty$ and to one at $+\infty$, and satisfies

$$\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a).$$

For a variable with density, $F_X(x) = \int_{-\infty}^x f_X(u) du$; where this function is differentiable, $F'_X(x) = f_X(x)$.

Example 2.2 (Waiting for a request). Let $\lambda = 0.2$ per minute. Then

$$\mathbb{P}(T > 10) = e^{-2} \approx 0.1353, \quad \mathbb{P}(2 < T \leq 5) = e^{-0.4} - e^{-1} \approx 0.3024.$$

The q -quantile, the time by which an arrival occurs with probability $q \in (0, 1)$, is

$$t_q = -\frac{\log(1 - q)}{\lambda}.$$

In particular, $t_{0.95} \approx 14.98$ minutes.

2.1.1 Memorylessness and its limits

For $s, t \geq 0$,

$$\mathbb{P}(T > s + t \mid T > s) = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = \mathbb{P}(T > t).$$

After waiting s units of time without an arrival, the remaining waiting time has the original distribution. This is *memorylessness*. It is an assumption about the mechanism: equipment that ages, scheduled arrivals, and demand that changes throughout the day need different models.

If $c > 0$, then $cT \sim \text{Exp}(\lambda/c)$, since $\mathbb{P}(cT > t) = e^{-(\lambda/c)t}$. Changing the time unit changes the numerical rate.

2.2 Expectation and functions of a random variable

Definition 2.3 (Expectation). For a discrete random variable with probability mass function (PMF) $p_X(x) = \mathbb{P}(X = x)$, define

$$\mathbb{E}[X] = \sum_x x p_X(x).$$

For a random variable with density f_X , define

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

A finite expectation is guaranteed when $\mathbb{E}[|X|] < \infty$, meaning that the corresponding sum or integral of absolute values is finite.

A PMF consists of nonnegative masses summing to one. A density consists of nonnegative heights integrating to one. Expectation is a probability-weighted average in either case. It need not be a possible observation: a fair six-sided die has expectation 3.5.

For $T \sim \text{Exp}(\lambda)$, integration by parts gives

$$\begin{aligned} \mathbb{E}[T] &= \int_0^{\infty} t \lambda e^{-\lambda t} dt \\ &= \left[-te^{-\lambda t} \right]_0^{\infty} + \int_0^{\infty} e^{-\lambda t} dt = \frac{1}{\lambda}. \end{aligned}$$

Thus a rate of 0.2 per minute corresponds to a mean waiting time of five minutes. The rate and the mean waiting time are reciprocals.

2.2.1 Computing the expectation of a function

For a measurable function g , provided $\mathbb{E}[|g(X)|] < \infty$,

$$\mathbb{E}[g(X)] = \begin{cases} \sum g(x)p_X(x), & X \text{ discrete,} \\ \int_{-\infty}^{\infty} g(x)f_X(x)dx, & X \text{ has a density.} \end{cases} \quad (2.1)$$

There is no need to find the distribution of $g(X)$ first.

Example 2.4 (A nonlinear function of waiting time). Suppose a waiting time $T \sim \text{Exp}(\lambda)$ receives a discount factor e^{-aT} , where $a > 0$. Then

$$\mathbb{E}[e^{-aT}] = \int_0^{\infty} e^{-at} \lambda e^{-\lambda t} dt = \frac{\lambda}{\lambda + a}.$$

This differs from $e^{-a\mathbb{E}[T]} = e^{-a/\lambda}$. In general, $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$.

Taking $g(x) = x^r$ gives the r -th raw moment. For a positive integer r , repeated integration by parts yields

$$\mathbb{E}[T^r] = \frac{r}{\lambda} \mathbb{E}[T^{r-1}] = \frac{r!}{\lambda^r}.$$

In particular, $\mathbb{E}[T^2] = 2/\lambda^2$.

2.2.2 Linearity

Theorem 2.5 (Linearity of expectation). *Let $X_1, \dots, X_n$ have finite expectations. For any constants a_i and b ,*

$$\mathbb{E}\left[b + \sum_{i=1}^n a_i X_i\right] = b + \sum_{i=1}^n a_i \mathbb{E}[X_i].$$

Independence is not required.

For a finite sample space, the reason is the distributive law:

$$\sum_{\omega \in \Omega} (aX(\omega) + bY(\omega))\mathbb{P}(\{\omega\}) = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

The same argument extends to integrable random variables using linearity of integration.

For an event A , its indicator $\mathbf{1}_A$ equals one on A and zero otherwise, so $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$. Therefore the expected number of events occurring among $A_1, \dots, A_n$ is

$$\mathbb{E}\left[\sum_{i=1}^n \mathbf{1}_{A_i}\right] = \sum_{i=1}^n \mathbb{P}(A_i),$$

even if the events depend on one another.

Linearity does not extend to arbitrary operations. For example, $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ holds for independent integrable X, Y , but need not hold for dependent variables.

2.3 Variance and standard deviation

Definition 2.6 (Variance). If $\mathbb{E}[X^2] < \infty$, the variance and standard deviation of X are

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2], \quad \text{sd}(X) = \sqrt{\text{Var}(X)}.$$

Variance measures squared spread around the mean. It is nonnegative, and equals zero exactly when X equals a constant with probability one. If X is measured in seconds, its variance is in seconds squared, while its standard deviation is in seconds.

Expanding the square and applying linearity gives the computational identity

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2. \quad (2.2)$$

For exponential waiting times,

$$\text{Var}(T) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}, \quad \text{sd}(T) = \frac{1}{\lambda}.$$

The mean and standard deviation are equal: exponential waiting times have substantial variation relative to their average.

2.3.1 Shifting, scaling, and adding

For constants a, b ,

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b, \quad \text{Var}(aX + b) = a^2 \text{Var}(X).$$

A shift changes the mean but leaves the spread unchanged. Multiplication by a changes standard deviation by $|a|$, and variance by a^2 . Variance is not a linear operation.

For random variables with finite second moments, define

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

Expanding the square of the sum gives

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y). \quad (2.3)$$

If X, Y are independent, their covariance is zero. Consequently, for independent $X_1, \dots, X_n$,

$$\text{Var}\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 \text{Var}(X_i).$$

Pairwise zero covariance is sufficient; it does not in general imply independence.

Example 2.7 (Why the independence condition matters). Suppose X and Y each have variance one. If they are independent, $\text{Var}(X + Y) = 2$. If $Y = X$, the variance is four. If $Y = -X$, it is zero. Knowing the individual variances alone does not determine the variance of the sum.

2.4 Poisson counts

Let $W_1, W_2, \dots$ be independent $\text{Exp}(\lambda)$ waiting times, and write

$$T_0 = 0, \quad T_k = W_1 + \dots + W_k.$$

The T_k are arrival times. The number of arrivals during $(0, t]$ is

$$N(t) = \max\{k \geq 0 : T_k \leq t\}.$$

This construction gives a *homogeneous Poisson process*: counts in disjoint intervals are independent, and the count distribution depends only on interval length. In particular, $N(t) \sim \text{Poisson}(\lambda t)$.

Definition 2.8 (Poisson distribution). A count $N \sim \text{Poisson}(m)$, with $m > 0$, has PMF

$$\mathbb{P}(N = k) = e^{-m} \frac{m^k}{k!}, \quad k = 0, 1, 2, \dots$$

The parameter m is the expected count in the observation interval. For a process with rate λ observed for time t , $m = \lambda t$.

Normalization follows from the exponential series:

$$\sum_{k=0}^{\infty} e^{-m} \frac{m^k}{k!} = e^{-m} e^m = 1.$$

Shifting the summation index gives the mean,

$$\mathbb{E}[N] = \sum_{k=1}^{\infty} k e^{-m} \frac{m^k}{k!} = m \sum_{j=0}^{\infty} e^{-m} \frac{m^j}{j!} = m.$$

The same method gives $\mathbb{E}[N(N-1)] = m^2$. Since $N^2 = N(N-1) + N$,

$$\mathbb{E}[N^2] = m^2 + m, \quad \text{Var}(N) = m.$$

The equality of mean and variance is a specific property of the Poisson model, not a general property of count data.

Example 2.9 (Counts and waiting times in the same service). With rate $\lambda = 0.2$ per minute, the ten-minute count has $m = 2$. Thus

$$\mathbb{P}(N(10) = 0) = e^{-2}, \quad \mathbb{P}(N(10) \geq 2) = 1 - e^{-2}(1 + 2) \approx 0.5940.$$

The equality $\mathbb{P}(N(10) = 0) = \mathbb{P}(T_1 > 10)$ expresses the same event using a count or a waiting time.

Independent Poisson counts add:

$$N_1 \sim \text{Poisson}(m_1), \quad N_2 \sim \text{Poisson}(m_2) \implies N_1 + N_2 \sim \text{Poisson}(m_1 + m_2).$$

Indeed, summing over $N_1 = j$ and using the binomial theorem gives

$$\mathbb{P}(N_1 + N_2 = k) = \frac{e^{-(m_1+m_2)}}{k!} \sum_{j=0}^k \binom{k}{j} m_1^j m_2^{k-j} = \frac{e^{-(m_1+m_2)} (m_1 + m_2)^k}{k!}.$$

Bursts, dependence between arrivals, or varying rates can make a single homogeneous Poisson model unsuitable.

2.5 Gamma waiting times and the event-count connection

The time T_k until the k -th arrival is a sum of k independent exponential variables. For $k = 2$, its density is obtained by convolution:

$$f_{T_2}(t) = \int_0^t \lambda e^{-\lambda u} \lambda e^{-\lambda(t-u)} du = \lambda^2 t e^{-\lambda t}, \quad t \geq 0.$$

Here convolution integrates over all ways the two waiting times can add to t . Repeating the argument gives

$$f_{T_k}(t) = \frac{\lambda^k}{(k-1)!} t^{k-1} e^{-\lambda t}, \quad t > 0.$$

Definition 2.10 (Gamma distribution: shape and rate). For $\alpha, \beta > 0$, the distribution $X \sim \text{Gamma}(\alpha, \beta)$ has density

$$f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0,$$

and zero density for $x < 0$. The gamma function is

$$\Gamma(\alpha) = \int_0^\infty u^{\alpha-1} e^{-u} du.$$

The parameters α and β are the *shape* and *rate*, respectively.

The value of a density at the single point zero is irrelevant to its probabilities; for $\alpha < 1$, this density diverges as $x \downarrow 0$ but remains integrable. Integration by parts gives $\Gamma(\alpha + 1) = \alpha \Gamma(\alpha)$, and $\Gamma(1) = 1$. Thus $\Gamma(k) = (k-1)!$ for positive integers k . The exponential is the case $\alpha = 1$, and

$$T_k \sim \text{Gamma}(k, \lambda).$$

The density is also well-defined for noninteger α , although the literal k -th-arrival interpretation requires an integer shape.

Using the substitution $u = \beta x$, we obtain

$$\mathbb{E}[X^r] = \frac{\Gamma(\alpha + r)}{\Gamma(\alpha) \beta^r}, \quad r > -\alpha.$$

Taking $r = 1, 2$ yields

$$\mathbb{E}[X] = \frac{\alpha}{\beta}, \quad \mathbb{E}[X^2] = \frac{\alpha(\alpha + 1)}{\beta^2}, \quad \text{Var}(X) = \frac{\alpha}{\beta^2}.$$

For integer shape these results also follow by adding the means and variances of independent exponential waiting times.

Warning (Rate versus scale). Some sources use $\text{Gamma}(\alpha, \theta)$ with *scale* $\theta = 1/\beta$. Then the mean is $\alpha\theta$ and the variance is $\alpha\theta^2$. Always check the density, rather than inferring the convention from the symbol used for the second parameter.

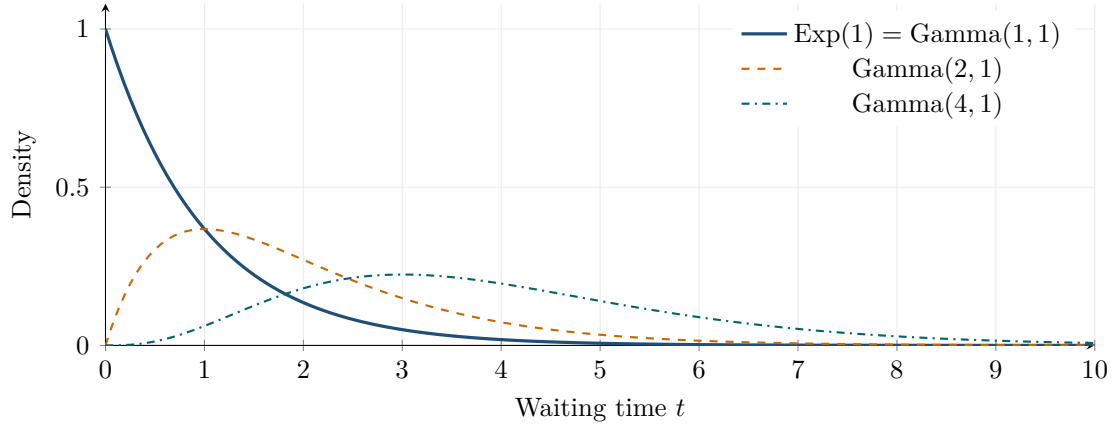


Figure 2.1: Exponential waiting times and Gamma sums of waiting times. All curves use rate one. Increasing the integer shape moves the mean from 1 to k , while the standard deviation grows only as $\sqrt{k}$.

2.5.1 Two descriptions of the same event

For $k \geq 1$,

$$\{T_k \leq t\} = \{N(t) \geq k\}. \quad (2.4)$$

Having waited long enough for the k -th arrival means having counted at least k arrivals. Consequently,

$$\mathbb{P}(T_k \leq t) = 1 - e^{-\lambda t} \sum_{j=0}^{k-1} \frac{(\lambda t)^j}{j!}.$$

Differentiating this expression recovers the Gamma density above. Equivalently, subtracting the CDFs of T_k and T_{k+1} gives the Poisson probability $\mathbb{P}(N(t) = k)$.

With rate 0.2 per minute, the waiting time for three requests has mean 15 minutes and variance 75 minutes squared. The probability of receiving the third request within ten minutes is

$$\mathbb{P}(T_3 \leq 10) = \mathbb{P}(N(10) \geq 3) = 1 - e^{-2}(1 + 2 + 2^2/2) \approx 0.3233.$$

2.6 Bernoulli and Binomial distributions

2.6.1 One binary outcome

If an event A occurs with probability p , its indicator $B = \mathbf{1}_A$ follows a *Bernoulli distribution*, $B \sim \text{Bernoulli}(p)$, where $0 \leq p \leq 1$:

$$\mathbb{P}(B = 1) = p, \quad \mathbb{P}(B = 0) = 1 - p.$$

Examples include a successful transmission, a user clicking a link, or an item failing inspection. Because $B^2 = B$,

$$\mathbb{E}[B] = p, \quad \mathbb{E}[B^2] = p, \quad \text{Var}(B) = p(1 - p).$$

Variance is largest at $p = 1/2$, and is zero when the outcome is certain.

2.6.2 Counting successes in a fixed number of trials

Let $B_1, \dots, B_n$ be independent Bernoulli variables with the same success probability p . Their sum

$$S_n = B_1 + \dots + B_n \sim \text{Binomial}(n, p)$$

counts the successes. For $0 < p < 1$,

$$\mathbb{P}(S_n = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, \dots, n.$$

There are $\binom{n}{k} = n!/[k!(n-k)!]$ ways to locate the k successes, and each such sequence has probability $p^k(1-p)^{n-k}$. At $p = 0$ or $p = 1$, the count is deterministically zero or n .

Linearity and independence give

$$\mathbb{E}[S_n] = np, \quad \text{Var}(S_n) = np(1-p).$$

For the proportion $\hat{p} = S_n/n$,

$$\mathbb{E}[\hat{p}] = p, \quad \text{Var}(\hat{p}) = \frac{p(1-p)}{n}.$$

For these independent, identically distributed trials, averaging divides the variance by n . Dependence can change this conclusion quantitatively.

Example 2.11 (Clicks among impressions). Under an independent model with $n = 100$ impressions and click probability $p = 0.03$, the click count has mean 3 and variance 2.91. The probability of no clicks is

$$\mathbb{P}(S_{100} = 0) = 0.97^{100} \approx 0.0476.$$

Repeated impressions to the same user need not be independent; different users need not have the same click probability.

2.6.3 The rare-event limit

Let $p_n = m/n$, with $m > 0$ fixed. For each fixed nonnegative integer k ,

$$\begin{aligned} \mathbb{P}(S_n = k) &= \binom{n}{k} \left(\frac{m}{n}\right)^k \left(1 - \frac{m}{n}\right)^{n-k} \\ &\longrightarrow \frac{m^k}{k!} e^{-m}. \end{aligned}$$

Thus $\text{Binomial}(n, m/n)$ approaches $\text{Poisson}(m)$: many trials, each with a small success probability, produce a Poisson count. Their moments agree in the limit:

$$np_n = m, \quad np_n(1-p_n) \longrightarrow m.$$

In the click example, the Poisson approximation gives a no-click probability $e^{-3} \approx 0.0498$.

2.7 Geometric waiting times

Instead of fixing the number of trials, keep performing independent Bernoulli trials until the first success. Let G count the trials, including the successful one. For $0 < p \leq 1$,

$$G \sim \text{Geometric}(p), \quad G \in \{1, 2, \dots\}.$$

For $0 < p < 1$,

$$\begin{aligned} \mathbb{P}(G = k) &= (1 - p)^{k-1}p, & k = 1, 2, \dots, \\ \mathbb{P}(G > k) &= (1 - p)^k, & k = 0, 1, \dots \end{aligned}$$

For $p = 1$, $G = 1$ with probability one. The alternative convention counts failures, $H = G - 1$, and has support $\{0, 1, \dots\}$.

Put $q = 1 - p$. Differentiating the geometric series $\sum_{k=0}^{\infty} q^k = (1 - q)^{-1}$, valid for $|q| < 1$, gives

$$\sum_{k=1}^{\infty} kq^{k-1} = \frac{1}{(1 - q)^2}, \quad \sum_{k=2}^{\infty} k(k - 1)q^{k-2} = \frac{2}{(1 - q)^3}.$$

Therefore

$$\mathbb{E}[G] = \frac{1}{p}, \quad \mathbb{E}[G(G - 1)] = \frac{2(1 - p)}{p^2}, \quad \text{Var}(G) = \frac{1 - p}{p^2}.$$

The failures count H has mean $(1 - p)/p$ and the same variance.

Memorylessness follows directly from the tail probabilities:

$$\mathbb{P}(G > m + k \mid G > m) = (1 - p)^k = \mathbb{P}(G > k).$$

For $0 < p < 1$, this holds for all nonnegative integers m, k . This is the discrete counterpart of exponential memorylessness.

2.7.1 From trials to continuous time

Suppose each trial lasts Δ time units and succeeds with probability $p_{\Delta} = \lambda\Delta$, where $0 < \lambda\Delta < 1$. Then

$$\mathbb{P}(\Delta G > t) = (1 - \lambda\Delta)^{\lfloor t/\Delta \rfloor} \longrightarrow e^{-\lambda t} \quad (\Delta \downarrow 0).$$

The rescaled geometric waiting time approaches $\text{Exp}(\lambda)$. Its moments also converge:

$$\mathbb{E}[\Delta G] = \frac{1}{\lambda}, \quad \text{Var}(\Delta G) = \frac{1 - \lambda\Delta}{\lambda^2} \longrightarrow \frac{1}{\lambda^2}.$$

The factor Δ is essential: a number of trials becomes a time only after multiplication by the duration of a trial.

Example 2.12 (A simple scrolling model). If a user independently stops after each video with probability $p = 0.1$, the number of videos viewed has mean 10, variance 90, and $\mathbb{P}(G > 20) = 0.9^{20} \approx 0.1216$. If stopping becomes more likely as the user gets tired, memorylessness no longer holds.

2.8 The Gaussian distribution

Definition 2.13 (Gaussian or normal distribution). For $\mu \in \mathbb{R}$ and $\sigma > 0$, the notation $X \sim \mathcal{N}(\mu, \sigma^2)$ means

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

The mean is μ , and the variance is σ^2 .

The Gaussian is symmetric around μ . Its support is the entire real line, so it can assign probability to negative values even when the observed quantity, such as a duration, cannot be negative.

2.8.1 Normalization and moments

Let $I = \int_{-\infty}^{\infty} e^{-z^2/2} dz$. Squaring the integral and switching to polar coordinates gives

$$I^2 = \int_{\mathbb{R}^2} e^{-(x^2+y^2)/2} dx dy = \int_0^{2\pi} \int_0^{\infty} e^{-r^2/2} r dr d\theta = 2\pi.$$

This explains the normalizing constant $\sqrt{2\pi}$. For $Z \sim \mathcal{N}(0, 1)$, write $\phi(z) = (2\pi)^{-1/2} e^{-z^2/2}$. Symmetry gives $\mathbb{E}[Z] = 0$. Since $\phi'(z) = -z\phi(z)$, integration by parts yields

$$\mathbb{E}[Z^2] = - \int_{-\infty}^{\infty} z\phi'(z) dz = [-z\phi(z)]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \phi(z) dz = 1.$$

Writing $X = \mu + \sigma Z$ now gives

$$\mathbb{E}[X] = \mu, \quad \text{Var}(X) = \sigma^2.$$

The second argument of $\mathcal{N}(\mu, \sigma^2)$ is variance, not standard deviation.

2.8.2 Standardization and interval probabilities

The transformation

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$$

converts a Gaussian variable to the standard normal. Define

$$\Phi(z) = \int_{-\infty}^z \phi(u) du.$$

Then

$$\mathbb{P}(a < X \leq b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right).$$

The density ϕ has no elementary antiderivative, so Φ is evaluated numerically. Symmetry gives $\Phi(-z) = 1 - \Phi(z)$, and

$$\mathbb{P}(|X - \mu| \leq \sigma) \approx 0.6827, \quad \mathbb{P}(|X - \mu| \leq 1.96\sigma) \approx 0.95.$$

These are probabilities under the assumed distribution, not a claim that arbitrary data obey the same coverage.

Example 2.14 (Measurement error). Suppose a sensor error $E \sim \mathcal{N}(0, 0.2^2)$, measured in the sensor's units. The probability of an absolute error larger than 0.4 is

$$\mathbb{P}(|E| > 0.4) = 2[1 - \Phi(2)] \approx 0.0455.$$

The central 95% interval is approximately $[-0.392, 0.392]$.

2.8.3 Transformations and sums

If $a \neq 0$, then

$$aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2).$$

If $a = 0$, the result is the constant b . For independent Gaussian variables,

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2),$$

their convolution gives

$$X + Y \sim \mathcal{N}(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2).$$

This conclusion concerns the entire distribution. Linearity of expectation and additivity of variance identify its parameters, but by themselves do not prove that the sum is Gaussian.

Many small independent additive effects can produce an approximately Gaussian aggregate. More precisely, for independent identically distributed variables with mean m and finite, nonzero variance v , the central limit theorem states

$$\frac{\sum_{i=1}^n X_i - nm}{\sqrt{nv}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Here convergence in distribution means convergence of CDF values at continuity points of the limiting CDF. A later chapter develops the theorem and its limitations. The finite-variance and dependence assumptions matter; adding arbitrary variables need not produce a Gaussian.

This explains the following useful connections:

$$\begin{aligned} \text{Binomial}(n, p) &\approx \mathcal{N}(np, np(1-p)), \\ \text{Poisson}(m) &\approx \mathcal{N}(m, m), \\ \text{Gamma}(k, \lambda) &\approx \mathcal{N}(k/\lambda, k/\lambda^2). \end{aligned}$$

The Binomial approximation needs both np and $n(1-p)$ to be large; the Poisson and Gamma approximations improve for large m and k . Near a support boundary or far in the tails, these approximations may be poor. For an integer-valued count, a continuity correction such as $\mathbb{P}(N \leq j) \approx \Phi((j + 1/2 - m)/\sqrt{m})$ accounts for the width of the probability mass at j .

2.9 Mixtures and heterogeneity

A population may contain several kinds of observations. A measurement can come from one of several instruments; a request can follow different execution paths; a randomly selected user can have a different activity rate from the next user. A mixture describes this two-stage mechanism.

2.9.1 First choose a component, then an observation

Let $Z \in \{1, \dots, K\}$ be a component label with

$$\mathbb{P}(Z = k) = \pi_k, \quad \pi_k \geq 0, \quad \sum_{k=1}^K \pi_k = 1.$$

Given $Z = k$, generate X with density f_k . Conditional notation $X | Z = k \sim f_k$ specifies the distribution inside component k . The law of total probability gives the marginal density

$$f_X(x) = \sum_{k=1}^K \pi_k f_k(x). \quad (2.5)$$

For discrete components, the identical formula applies to their PMFs. The label Z may be unobserved; it is then a *latent variable*.

The weights average *distributions*. They do not mean that each observation is the weighted sum of one draw from every component. For example, choosing X_1 or X_2 with equal probability gives a mixture, whereas $(X_1 + X_2)/2$ averages two observations. Those operations usually produce different distributions.

2.9.2 Mean and variance of a finite mixture

Let component k have mean μ_k and variance σ_k^2 . Assume finite second moments. Integrating (2.5) gives

$$\mu = \mathbb{E}[X] = \sum_{k=1}^K \pi_k \mu_k, \quad \mathbb{E}[X^2] = \sum_{k=1}^K \pi_k (\sigma_k^2 + \mu_k^2).$$

Subtracting μ^2 and regrouping terms gives

$$\text{Var}(X) = \underbrace{\sum_{k=1}^K \pi_k \sigma_k^2}_{\text{within-component variation}} + \underbrace{\sum_{k=1}^K \pi_k (\mu_k - \mu)^2}_{\text{between-component variation}}. \quad (2.6)$$

The spread of a pooled population includes variation within its groups and differences between group means.

In conditional notation these identities are the laws of total expectation and total variance:

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X | Z]], \quad (2.7)$$

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X | Z)] + \text{Var}(\mathbb{E}[X | Z]). \quad (2.8)$$

Here $\mathbb{E}[X | Z]$ is itself a random variable: its value is μ_k when $Z = k$. Likewise, $\text{Var}(X | Z) = \sigma_k^2$. The outer expectation averages over the component label.

Example 2.15 (Two sensor settings). Choose either setting with probability $1/2$, and suppose

$$X | Z = 1 \sim \mathcal{N}(-2, 1), \quad X | Z = 2 \sim \mathcal{N}(2, 1).$$

The pooled mean is zero. Its variance is

$$\text{Var}(X) = \underbrace{1}_{\text{within settings}} + \underbrace{4}_{\text{between settings}} = 5.$$

The density has two peaks. The single Gaussian $\mathcal{N}(0, 5)$ has the same mean and variance but a different shape (Figure 2.2). For instance, the mixture gives

$$\mathbb{P}(|X| \leq 1) = \Phi(3) - \Phi(1) \approx 0.1573,$$

whereas $\mathcal{N}(0, 5)$ gives $2\Phi(1/\sqrt{5}) - 1 \approx 0.3453$. Matching two moments does not identify a distribution.

If instead $X_1 \sim \mathcal{N}(-2, 1)$ and $X_2 \sim \mathcal{N}(2, 1)$ are independent and we average them, then

$$\frac{X_1 + X_2}{2} \sim \mathcal{N}(0, 1/2).$$

This is neither the mixture nor the Gaussian that matches its moments.

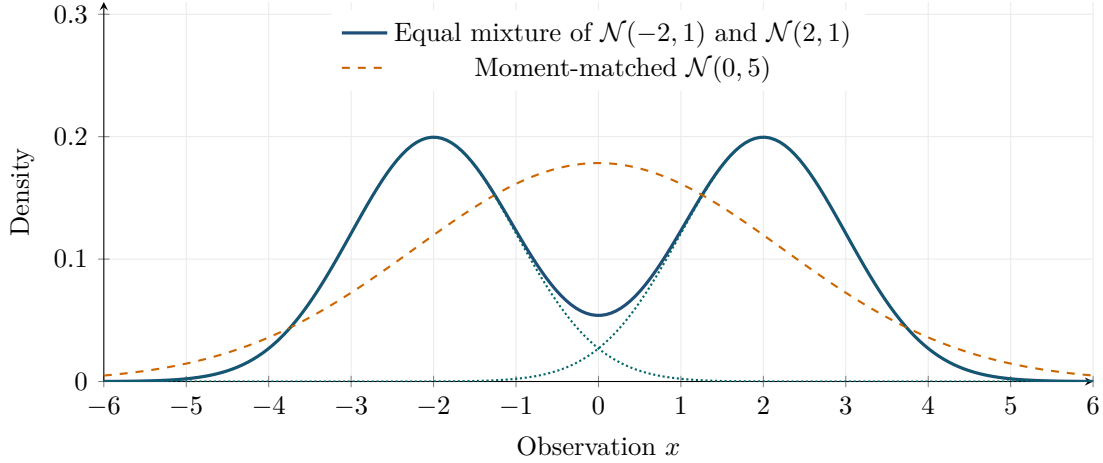


Figure 2.2: A Gaussian mixture and a Gaussian with the same mean and variance. The dotted curves show the weighted component densities; they add to the blue mixture curve. Equal first and second moments do not imply equal probabilities for an interval.

2.9.3 A continuous distribution of parameters

Heterogeneity need not consist of a few categories. Let a parameter Θ have density g , and generate X conditional on its value. The marginal density or PMF is

$$p_X(x) = \int p(x | \theta)g(\theta) d\theta.$$

For continuous X , $p(x | \theta)$ denotes a conditional density; for discrete X , it denotes a conditional probability mass. The integral averages over possible parameters. In general, replacing Θ by $\mathbb{E}[\Theta]$ inside $p(x | \Theta)$ does not produce the same distribution.

Equations (2.7) and (2.8) also hold for this continuous mixing parameter when the relevant moments exist. They allow moment calculations without evaluating the full mixture integral.

2.9.4 Different users have different event rates

Suppose the count N for a randomly chosen user over a fixed observation window satisfies

$$N | \Lambda = \lambda \sim \text{Poisson}(\lambda).$$

Here Λ is the user's expected count for that window; all users are observed over the same duration. Conditional means and variances are both Λ , hence

$$\mathbb{E}[N] = \mathbb{E}[\Lambda], \quad \text{Var}(N) = \mathbb{E}[\Lambda] + \text{Var}(\Lambda).$$

If users differ in their rates, the marginal variance exceeds the mean. This is *overdispersion* relative to a Poisson model. It can occur even when each individual user's count is Poisson. Other mechanisms, such as dependent arrivals, can also cause overdispersion.

Take $\Lambda \sim \text{Gamma}(\alpha, \beta)$, in the rate convention. Combining two distributions already studied gives

$$\mathbb{E}[N] = \frac{\alpha}{\beta}, \quad \text{Var}(N) = \frac{\alpha}{\beta} + \frac{\alpha}{\beta^2}.$$

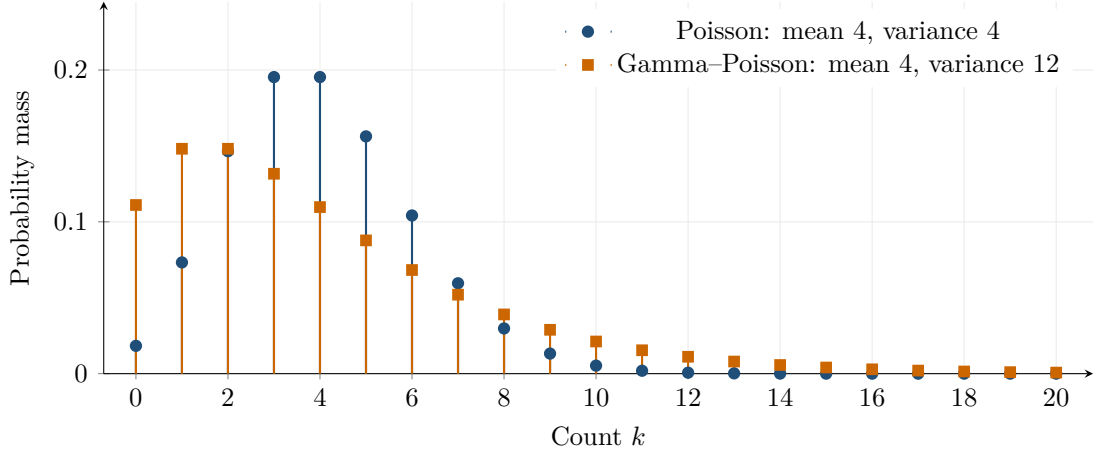


Figure 2.3: Count distributions with the same mean, four. Mixing Poisson rates with a Gamma(2, 1/2) distribution produces more zeros and a longer upper tail; its variance is twelve. The displayed values are probability masses, not a continuous density.

For completeness, the mixture PMF follows from the gamma integral:

$$\begin{aligned}\mathbb{P}(N = k) &= \int_0^\infty \frac{e^{-\lambda} \lambda^k}{k!} \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda} d\lambda \\ &= \frac{\Gamma(k + \alpha)}{\Gamma(\alpha)k!} \left(\frac{\beta}{\beta + 1} \right)^\alpha \left(\frac{1}{\beta + 1} \right)^k, \quad k \geq 0.\end{aligned}$$

This is a *negative binomial distribution*, in a convention that counts failures and permits any real shape $\alpha > 0$. Its construction, rather than its name, explains why it is useful for heterogeneous counts.

For $\alpha = 2, \beta = 1/2$, the mean is 4 and the variance is 12. A Poisson distribution with mean 4 has variance only 4. The heterogeneous model also gives

$$\mathbb{P}(N = 0) = \left(\frac{1}{3} \right)^2 = \frac{1}{9} \approx 0.1111,$$

compared with $e^{-4} \approx 0.0183$ under the single-rate Poisson model.

2.9.5 One random probability or a new one for every trial?

Let $P \in [0, 1]$ be a random success probability and $B \mid P = p \sim \text{Bernoulli}(p)$. Then

$$\mathbb{P}(B = 1) = \mathbb{E}[P],$$

so a single observation is still Bernoulli with parameter $q = \mathbb{E}[P]$. Heterogeneity cannot be identified from the marginal distribution of one binary observation alone.

For repeated trials, the sampling mechanism matters. If a user has one shared probability P and, conditional on it, $S \mid P \sim \text{Binomial}(n, P)$, then

$$\begin{aligned}\mathbb{E}[S] &= nq, \\ \text{Var}(S) &= n\mathbb{E}[P(1 - P)] + n^2 \text{Var}(P) \\ &= nq(1 - q) + n(n - 1) \text{Var}(P).\end{aligned}$$

If $n > 1$ and $\text{Var}(P) > 0$, the variance exceeds $nq(1-q)$. Trials are independent conditional on the user's probability, but generally dependent after pooling users with different probabilities.

If instead each trial independently draws a fresh P_i from the same distribution and then independently generates $B_i | P_i$, the unconditional B_i are independent Bernoulli(q), and their sum is Binomial(n, q). The choice of what is shared across observations is part of the model.

2.10 Extension: Gaussian scale mixtures and Student's t

Hidden groups may differ in spread even when they have the same mean. One simple *contaminated Gaussian* model is

$$X \sim (1 - \varepsilon)\mathcal{N}(\mu, \sigma^2) + \varepsilon\mathcal{N}(\mu, a^2\sigma^2), \quad 0 < \varepsilon < 1, \quad a > 1.$$

A fraction ε of observations comes from a noisier component. The mean remains μ , while

$$\text{Var}(X) = \sigma^2[(1 - \varepsilon) + \varepsilon a^2].$$

Even a small fraction can greatly increase the total variance.

A continuous version randomizes the *precision*, the inverse of variance. Let

$$\tau \sim \text{Gamma}(\nu/2, \nu/2), \quad Z | \tau \sim \mathcal{N}(0, \tau^{-1}), \quad \nu > 0.$$

Integrating out τ gives the standard Student distribution, $Z \sim t_\nu$, with density

$$f_Z(z) = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi} \Gamma(\nu/2)} \left(1 + \frac{z^2}{\nu}\right)^{-(\nu+1)/2}.$$

Small precisions correspond to occasional large conditional variances. The resulting tails decay as a power of $|z|$, much more slowly than Gaussian tails.

For $\nu > 1$, $\mathbb{E}[Z] = 0$; for $\nu > 2$,

$$\text{Var}(Z) = \mathbb{E}[\tau^{-1}] = \frac{\nu}{\nu - 2}.$$

For $1 < \nu \leq 2$, the mean exists but the second moment is infinite; for $0 < \nu \leq 1$, the mean does not exist. Symmetry alone does not guarantee a finite mean.

If $X = \mu + sZ$, where $s > 0$, then s is a scale parameter, not the standard deviation: when $\nu > 2$,

$$\text{Var}(X) = s^2 \frac{\nu}{\nu - 2}.$$

As ν grows, the standard Student distribution approaches $\mathcal{N}(0, 1)$. It connects Gamma variability, Gaussian conditional distributions, and heavy-tailed observations in a single model.

2.11 Distribution reference

Distribution	Support	$\mathbb{E}[X]$	$\text{Var}(X)$
$\text{Exp}(\lambda)$	$[0, \infty)$	$1/\lambda$	$1/\lambda^2$
$\text{Poisson}(m)$	$0, 1, \dots$	m	m
$\text{Gamma}(\alpha, \beta)$	$[0, \infty)$	α/β	α/β^2
$\text{Bernoulli}(p)$	$0, 1$	p	$p(1-p)$
$\text{Binomial}(n, p)$	$0, \dots, n$	np	$np(1-p)$
$\text{Geometric}(p)$	$1, 2, \dots$	$1/p$	$(1-p)/p^2$
$\mathcal{N}(\mu, \sigma^2)$	$\mathbb{R}$	μ	σ^2
t_ν	$\mathbb{R}$	$0 \ (\nu > 1)$	$\nu/(\nu-2) \ (\nu > 2)$

Here $\lambda, \alpha, \beta, m, \nu > 0$, $\sigma > 0$, and n is a nonnegative integer. The Bernoulli and Binomial parameters allow $0 \leq p \leq 1$; the Geometric requires $0 < p \leq 1$. Gamma uses shape and rate. Geometric counts trials including the first success. Student moments exist only under the conditions shown.

Connection	Required mechanism or condition
Exponential to Gamma	Sum independent waiting times with a common rate.
Exponential to Poisson	Count arrivals generated by independent exponential interarrival times.
Bernoulli to Binomial	Sum a fixed number of independent trials with the same success probability.
Binomial to Poisson	Take $n \rightarrow \infty$, $p \rightarrow 0$, with $np = m$.
Geometric to Exponential	Rescale trial duration to Δ , set $p = \lambda\Delta$, and let $\Delta \downarrow 0$.
Poisson to negative binomial	Mix the Poisson rate with a Gamma distribution.
Gaussian to Student	Mix the Gaussian precision with $\text{Gamma}(\nu/2, \nu/2)$.

2.12 Exercises

Exercise 2.1 (Masses, CDF, and moments). A discrete variable satisfies

$$p_X(-1) = a, \quad p_X(0) = 2a, \quad p_X(2) = a.$$

Find a , write its CDF, and compute $\mathbb{P}(-1 < X \leq 2)$, $\mathbb{E}[X]$, and $\text{Var}(X)$.

Exercise 2.2 (A density and a nonlinear expectation). Let $f_X(x) = cx$ for $0 \leq x \leq 1$, and zero otherwise. Find c , derive the CDF, and calculate $\mathbb{P}(0.2 < X \leq 0.7)$. Find $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and $\text{Var}(X)$. Explain why $\mathbb{E}[X^2] \neq \mathbb{E}[X]^2$.

Exercise 2.3 (One arrival process, three distributions). Requests follow a homogeneous Poisson process at rate 0.5 per minute. Find the probability of no requests in four minutes, the probability that the third request arrives within four minutes, and the mean and variance of the waiting time until the third request. Given that no request arrives in the first six minutes, find the probability of waiting at least two more minutes for the first request.

Exercise 2.4 (Expectation of a function). For $T \sim \text{Exp}(\lambda)$ and $a > 0$, find $\mathbb{E}[e^{-aT}]$ and $\text{Var}(e^{-aT})$. For $\lambda = a = 1$, compare $\mathbb{E}[e^{-T}]$ with $e^{-\mathbb{E}[T]}$.

Exercise 2.5 (Averages and dependence). Let $X_1, \dots, X_n$ be independent, with a common mean μ and variance σ^2 . Find the mean and variance of $\bar{X} = n^{-1} \sum_i X_i$. What changes if $X_1 = \dots = X_n = X$ with probability one?

Exercise 2.6 (Rare successes and waiting). Each of 200 independent trials succeeds with probability 0.01. Find the count's mean, variance, exact no-success probability, and its Poisson approximation. In an unlimited sequence of these trials, find the mean and variance of the number of trials until the first success.

Exercise 2.7 (Gaussian sums and standardization). Independent measurement errors satisfy $X \sim \mathcal{N}(1, 4)$ and $Y \sim \mathcal{N}(-1, 9)$. Find the distributions of $X + Y$ and $2X - Y$. Express $\mathbb{P}(|X + Y| \leq 2)$ using Φ .

Exercise 2.8 (Mixture versus average). Choose $X_1 \sim \mathcal{N}(-3, 1)$ with probability $1/4$ and $X_2 \sim \mathcal{N}(1, 1)$ otherwise; call the selected value M . Find $\mathbb{E}[M]$ and $\text{Var}(M)$. Compare them with the mean and variance of $A = (X_1 + 3X_2)/4$, when X_1, X_2 are independent.

Exercise 2.9 (Heterogeneous counts). For a fixed observation window, let $N | \Lambda \sim \text{Poisson}(\Lambda)$ and $\Lambda \sim \text{Gamma}(3, 1/2)$. Find $\mathbb{E}[N]$, $\text{Var}(N)$, and $\mathbb{P}(N = 0)$. Compare with a Poisson distribution having the same mean.

Exercise 2.10 (A probability shared across trials). A user has probability $P = 0.1$ or $P = 0.5$, each with probability $1/2$. Conditional on P , their ten binary responses are independent. Find the mean and variance of the success count. How does its variance change if a fresh independent P_i is drawn for each response instead?

Exercise 2.11 (Student scale). Let $Z \sim t_4$ and $X = 2 + 3Z$. Find the mean, variance, and standard deviation of X . What would fail if $Z \sim t_2$?

Short answers

1. $a = 1/4$, with CDF

$$F_X(x) = \begin{cases} 0, & x < -1, \\ 1/4, & -1 \leq x < 0, \\ 3/4, & 0 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$

Then $\mathbb{P}(-1 < X \leq 2) = 3/4$, $\mathbb{E}[X] = 1/4$, $\mathbb{E}[X^2] = 5/4$, and $\text{Var}(X) = 19/16$.

2. $c = 2$. The CDF is zero for $x < 0$, x^2 on $[0, 1]$, and one for $x > 1$. The interval probability is 0.45. Moreover, $\mathbb{E}[X] = 2/3$, $\mathbb{E}[X^2] = 1/2$, and $\text{Var}(X) = 1/18$. Squaring is nonlinear; the difference between the two expectations in question is the positive variance.
3. The no-request probability is $e^{-2} \approx 0.1353$; the third-arrival probability is $1 - 5e^{-2} \approx 0.3233$. The third-arrival time has distribution $\text{Gamma}(3, 0.5)$, mean 6 minutes and variance 12 minutes squared. Memorylessness gives $e^{-1} \approx 0.3679$ for the additional wait.
4. $\mathbb{E}[e^{-aT}] = \lambda/(\lambda + a)$ and

$$\text{Var}(e^{-aT}) = \frac{\lambda}{\lambda + 2a} - \left(\frac{\lambda}{\lambda + a} \right)^2.$$

At $\lambda = a = 1$, the expectation is $1/2$, the variance is $1/12$, and $e^{-\mathbb{E}[T]} = e^{-1} \neq 1/2$.

5. In both cases the mean is μ . The independent average has variance σ^2/n ; the identical-copy average equals X and has variance σ^2 .
6. The count has mean 2, variance 1.98, and no-success probability $0.99^{200} \approx 0.1340$; the approximation is $e^{-2} \approx 0.1353$. The Geometric waiting count has mean 100 and variance 9900.
7. $X + Y \sim \mathcal{N}(0, 13)$, $2X - Y \sim \mathcal{N}(3, 25)$, and the interval probability is $2\Phi(2/\sqrt{13}) - 1$.
8. $\mathbb{E}[M] = 0$, $\text{Var}(M) = 1 + 3 = 4$. The average has mean zero and variance $(1/4)^2 + (3/4)^2 = 5/8$; in fact $A \sim \mathcal{N}(0, 5/8)$.
9. The mean is 6, the variance is $6 + 12 = 18$, and $\mathbb{P}(N = 0) = (1/3)^3 = 1/27 \approx 0.0370$. A Poisson(6) variable has variance 6 and zero-count probability $e^{-6} \approx 0.00248$.
10. $q = \mathbb{E}[P] = 0.3$ and $\text{Var}(P) = 0.04$. The mean is 3, and the shared-probability variance is $10(0.3)(0.7) + 10(9)(0.04) = 5.7$. With fresh independent probabilities, the variance is 2.1.
11. $\mathbb{E}[X] = 2$, $\text{Var}(X) = 9 \cdot 4/(4 - 2) = 18$, and $\text{sd}(X) = 3\sqrt{2}$. For t_2 , the mean still exists but the variance is infinite.

Further reading

Distributions in Real Life, Yandex School of Data Analysis, 15 February 2024. The lecture develops the connections between event counts and waiting times, and applications of mixtures to heterogeneous populations.